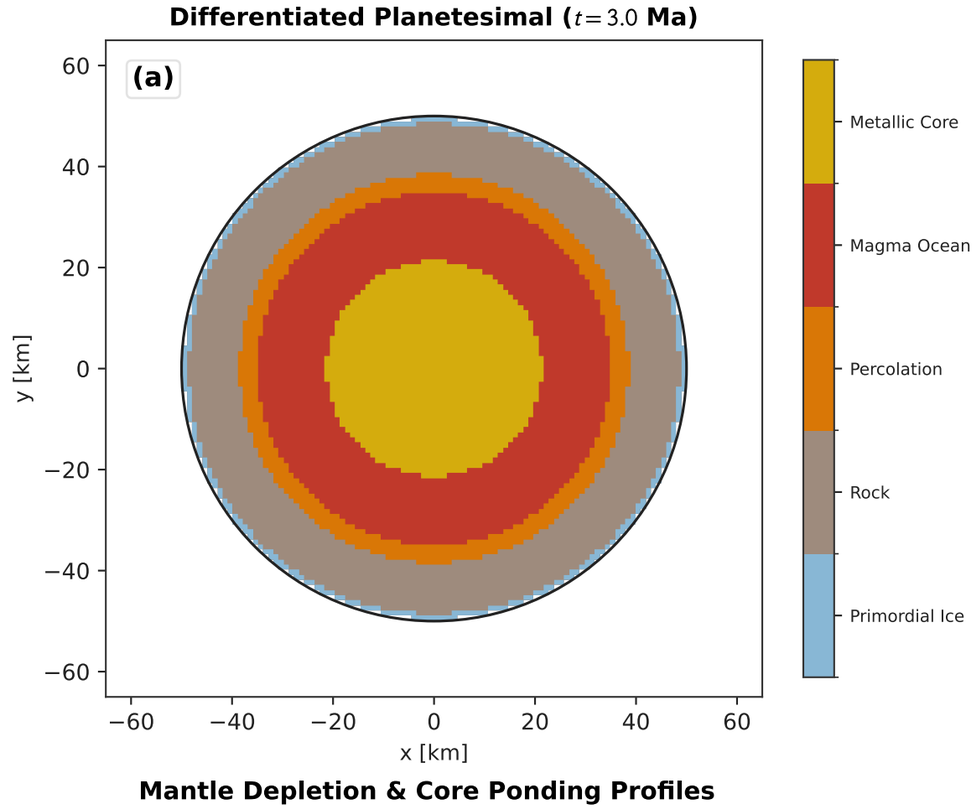
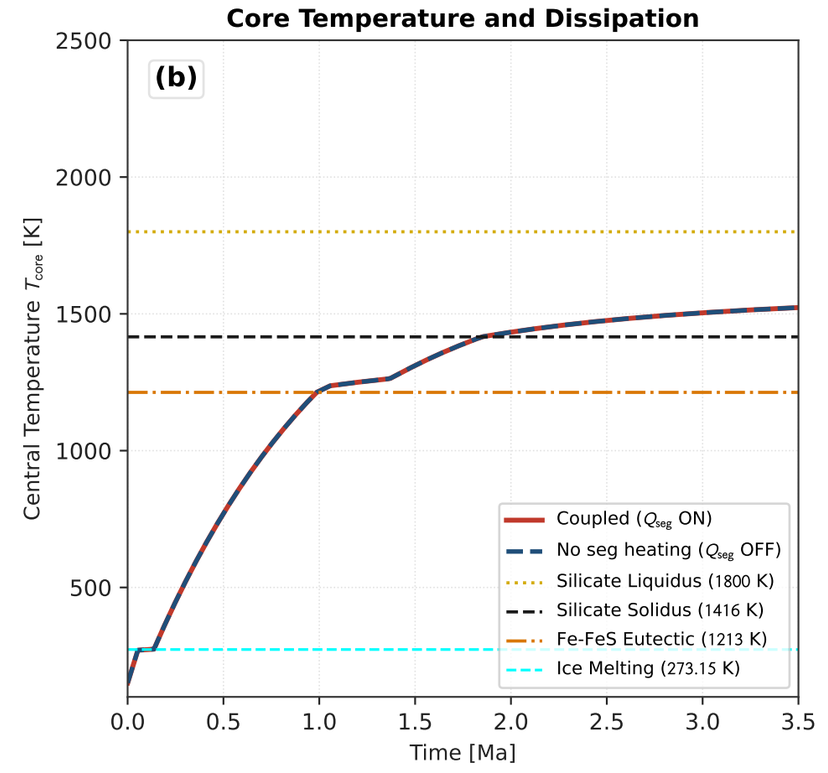
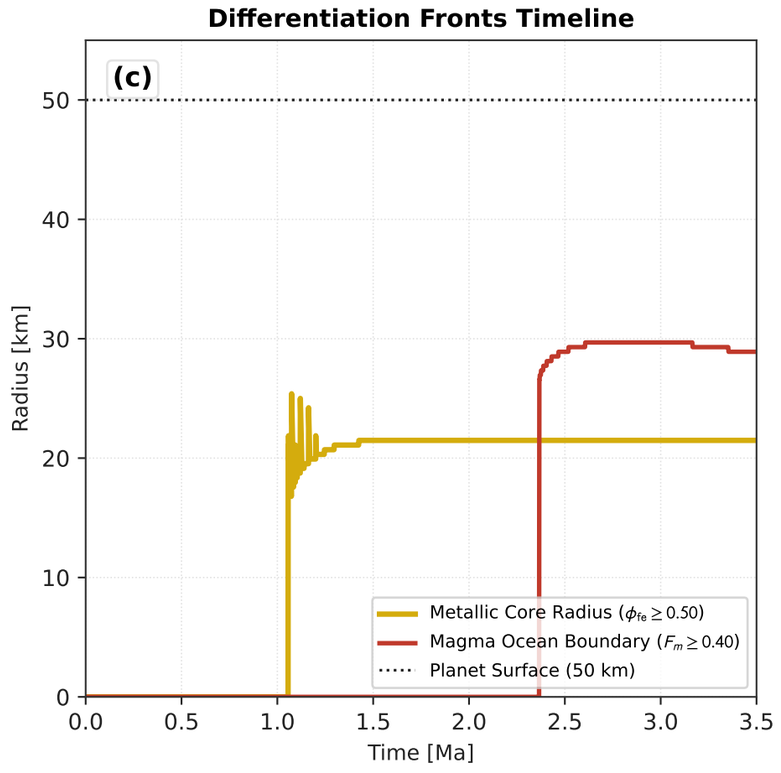
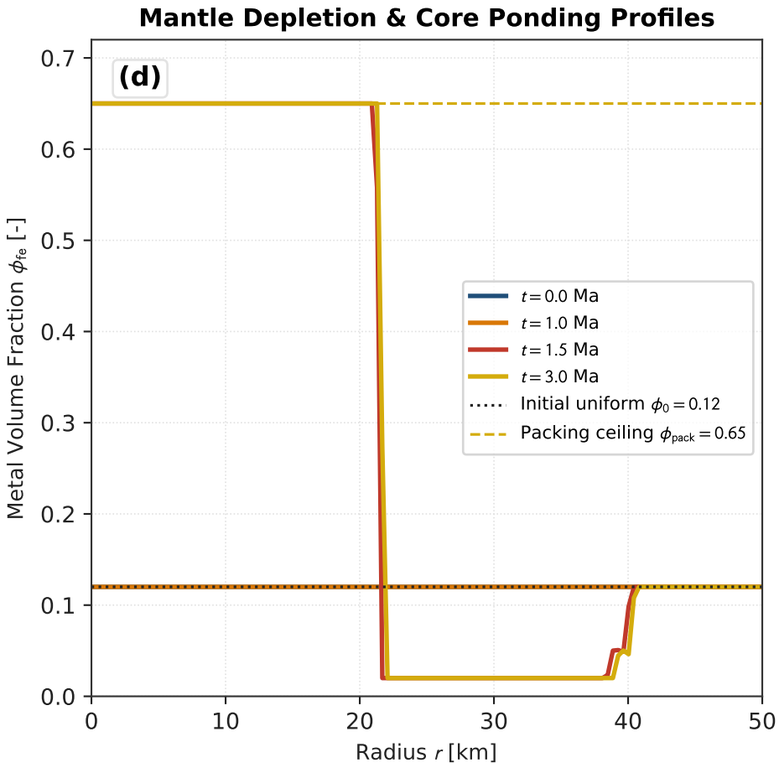
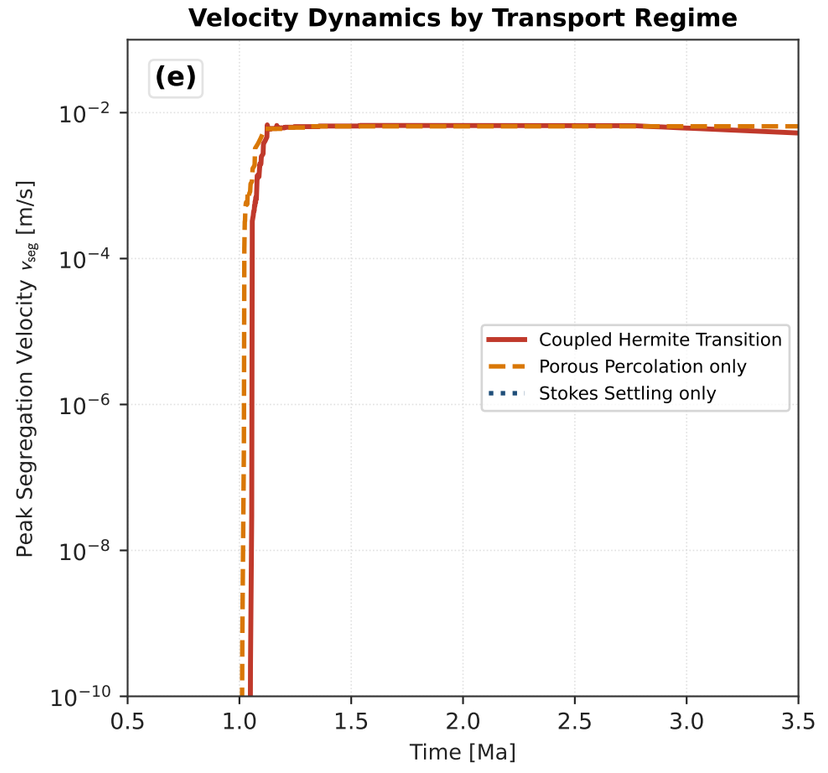
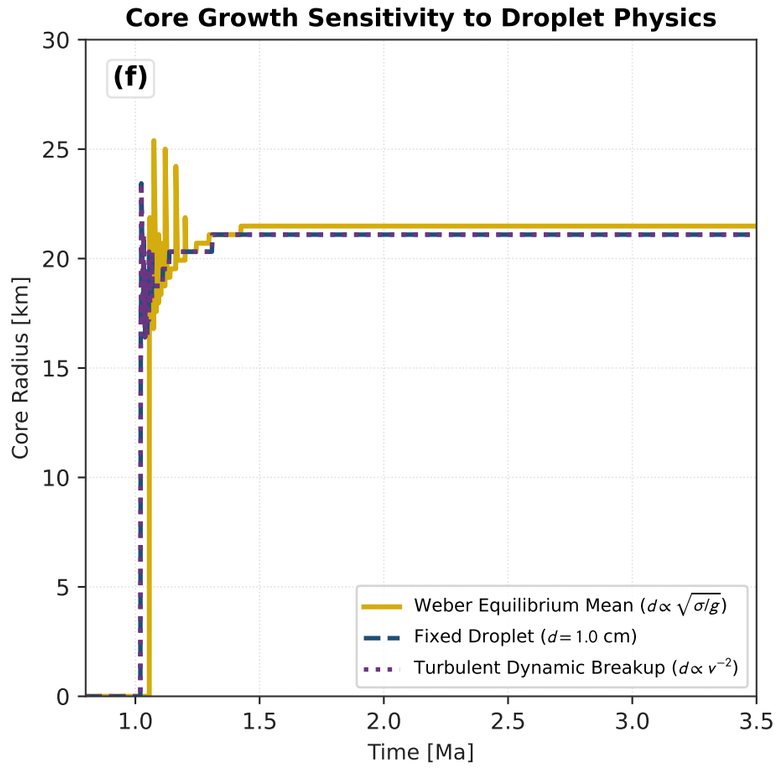

Differentiated Planetesimal ($t = 3.0$ Ma)
(a)
Metallic Core
Magma Ocean
Percolation
Rock
Primordial Ice
y [km]
x [km]


Core Temperature and Dissipation
(b)
Central Temperature T_{core} [K]
Time [Ma]
Coupled (Q_{seg} ON)
No seg heating (Q_{seg} OFF)
Silicate Liquidus (1800 K)
Silicate Solidus (1416 K)
Fe-FeS Eutectic (1213 K)
Ice Melting (273.15 K)


Differentiation Fronts Timeline
(c)
Radius [km]
Time [Ma]
Metallic Core Radius ($\phi_{fe} \geq 0.50$)
Magma Ocean Boundary ($F_m \geq 0.40$)
Planet Surface (50 km)


Mantle Depletion & Core Ponding Profiles
(d)
Metal Volume Fraction ϕ_{fe} [-]
Radius r [km]
$t = 0.0$ Ma
$t = 1.0$ Ma
$t = 1.5$ Ma
$t = 3.0$ Ma
Initial uniform $\phi_0 = 0.12$
Packing ceiling $\phi_{pack} = 0.65$


Velocity Dynamics by Transport Regime
(e)
Peak Segregation Velocity v_{seg} [m/s]
Time [Ma]
Coupled Hermite Transition
Porous Percolation only
Stokes Settling only


Core Growth Sensitivity to Droplet Physics
(f)
Core Radius [km]
Time [Ma]
Weber Equilibrium Mean ($d \propto \sqrt{\sigma/g}$)
Fixed Droplet ($d = 1.0$ cm)
Turbulent Dynamic Breakup ($d \propto v^{-2}$)